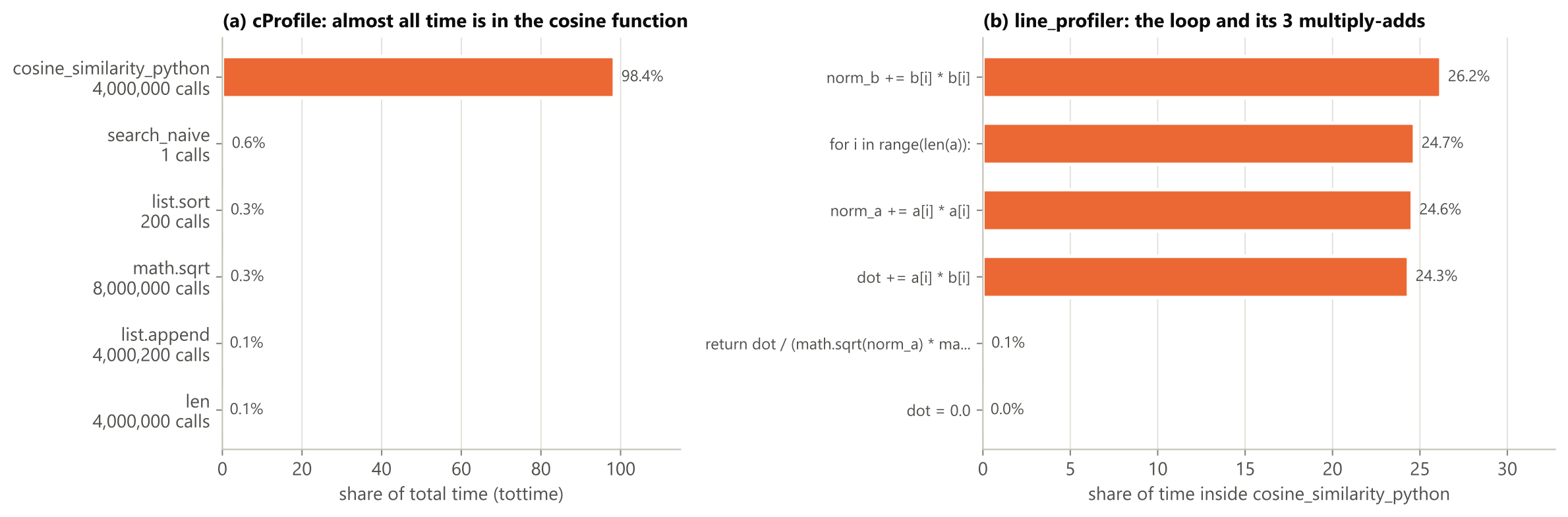
(a) cProfile: almost all time is in the cosine function
cosine_similarity_python
4,000,000 calls
98.4%
search_naive
1 calls
0.6%
list.sort
200 calls
0.3%
math.sqrt
8,000,000 calls
0.3%
list.append
4,000,200 calls
0.1%
len
4,000,000 calls
0.1%
0
20
40
60
80
100
share of total time (tottime)
(b) line_profiler: the loop and its 3 multiply-adds
norm_b += b[i] * b[i]
26.2%
for i in range(len(a)):
24.7%
norm_a += a[i] * a[i]
24.6%
dot += a[i] * b[i]
24.3%
return dot / (math.sqrt(norm_a) * ma...
0.1%
dot = 0.0
0.0%
0
5
10
15
20
25
30
share of time inside cosine_similarity_python